COMPOSITE TENSILE TESTING JIG

Documentation

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Background

Composite materials are widely used in Formula SAE (FSAE) vehicle structures due to their high specific strength and stiffness. In applications where mechanical fasteners or inserts are bonded into composite laminates, the ability of the structure to transfer loads normal to the laminate surface is a critical design consideration. Unlike metallic structures, composites are particularly sensitive to through-thickness loading and localized stress concentrations, making experimental validation essential.

To ensure structural integrity and compliance with FSAE regulations, composite inserts must demonstrate sufficient resistance to tensile pull-out loads. The composite pull-out test evaluates the load-bearing capacity of an insert bonded or embedded within a laminate when subjected to axial tension. This test provides validation of the laminate layup, insert design, bonding method, and manufacturing quality.

This report presents the composite pull-out testing conducted in accordance with FSAE requirements to verify insert strength and demonstrate regulatory compliance.

Project Overview

The objective of this project is to design a pull-out test jig with minimal deflection under a pull-out load of 15 kN, in order to ensure that measured deflection is primarily attributed to the composite plate, rather than the fixture. This is required as the experiment aims to investigate the force–deflection behaviour of the composite plate.

Objectives

- Design a testing jig for a composite flat panel that can be fitted into the instron
- Perform finite element analysis to ensure the testing jig doesn't undergo any deformation under load
- Document the entire design to manufacture process and any relevant regulated to the design of the jig or testing procedure

Rules & Compliance: ISO 527-1

5.1.2 Test Speeds: The tensile-testing machine shall be capable of maintaining the test speeds as specified in Table 1.

Table 1 — Recommended test speeds

Test speed v mm/min	Tolerance %
0,125	±20
0,25	
0,5	
1	
2	
5	
10	
20	±10
50	
100	
200	
300	
500	

7 Number of Test Specimen - 7.1: A minimum of five test specimens shall be tested for each of the required directions of testing. The number of measurements may be more than five if greater precision of the mean value is required. It is possible to evaluate this by means of the confidence interval (95 % probability, see ISO 2602).

8 Conditioning: The test specimen shall be conditioned as specified in the appropriate standard for the material concerned. In the absence of this information, the most appropriate set of conditions from ISO 291 shall be selected and the conditioning time is at least 16 h, unless otherwise agreed upon by the interested parties, for example, for testing at elevated or low temperatures.

The preferred atmosphere is (23 $\pm$ 2) $^{\circ}$ C and (50 $\pm$ 10) % R.H., except when the properties of the material are known to be insensitive to moisture, in which case humidity control is unnecessary.

9.1 Test Atmosphere: Conduct the test in the same atmosphere used for conditioning the test specimen, unless otherwise agreed upon by the interested parties, for example, for testing at elevated or low temperatures.

Assumptions

- Friction is negligible (Doesn't take load therefore, doesn't reduce stress incurred)
- The M8 bolt reaction forces of a structural steel plate are similar to that off a composite plate
- The jig's top plate takes all the M8 bolt reaction forces
- The simulated merged top plate and flanges have the same structural rigidity as the welded pieces

Project

Design

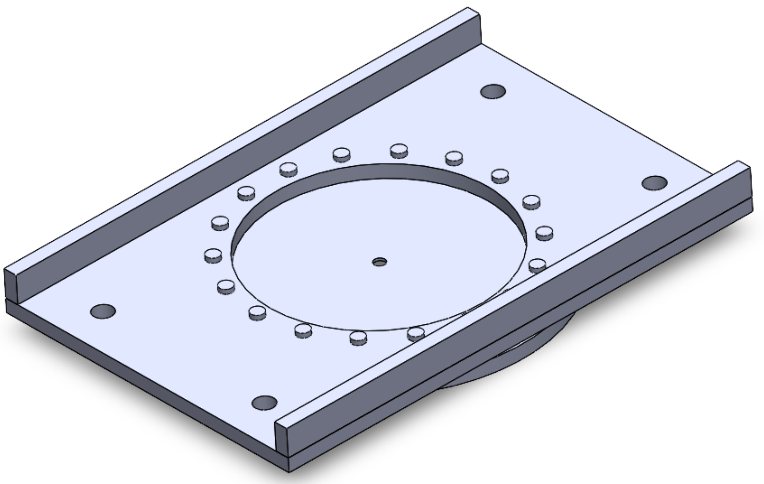


Figure 1: CAD Model of the Testing Jig (SolidWorks)

The design consists of eighteen M8 bolts configured in a circular pattern fixing a carbon plate between two 16mm steel plate and four M20 threaded rods fixing the jig to the

Instron. The flanges on the sides are two rectangular steel pieces that are welded on, to increase second moment of area to decrease unwanted plate deflection.

Finite Element Analysis – Pre Process

Carbon Plate Forces Approximation:

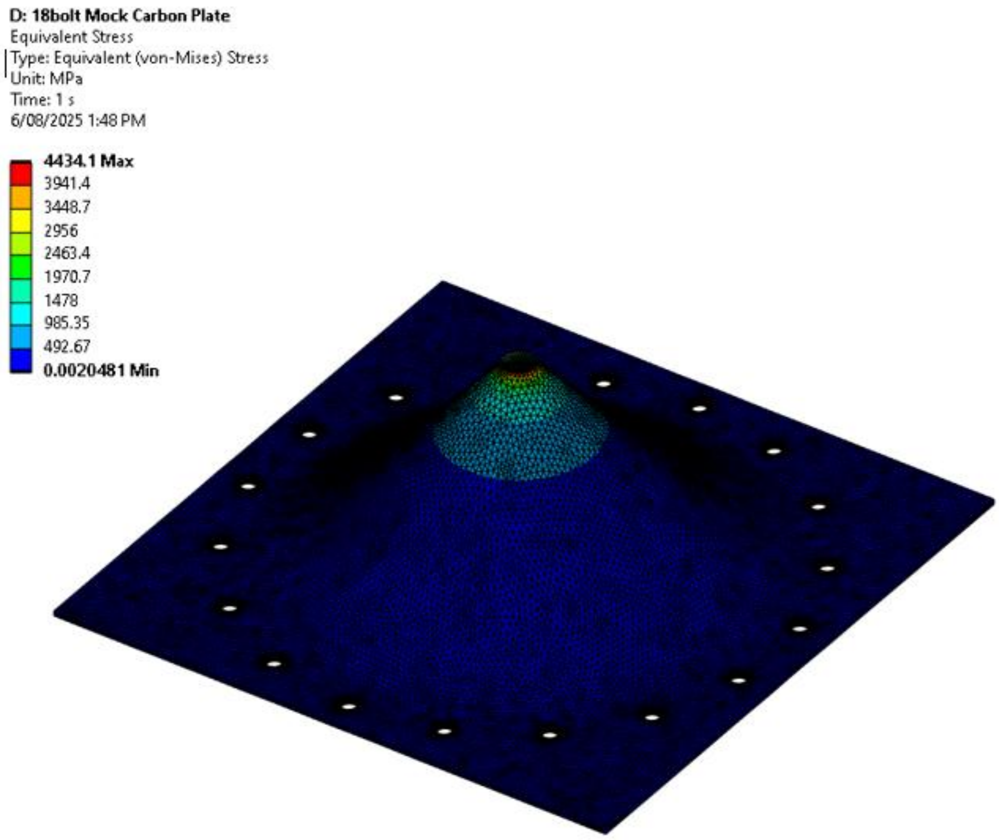


Figure 2: FEA of Mock Composite Plate

To figure out the boundary conditions to assign the FEA on the testing jig an initial FEA simulation was conducted on the carbon plate where the force reactions on the M8 holes shown in appendix (figure 2 shows the stress distribution across the plate) which is then inverted as shown in figure 3.

Boundary Conditions:

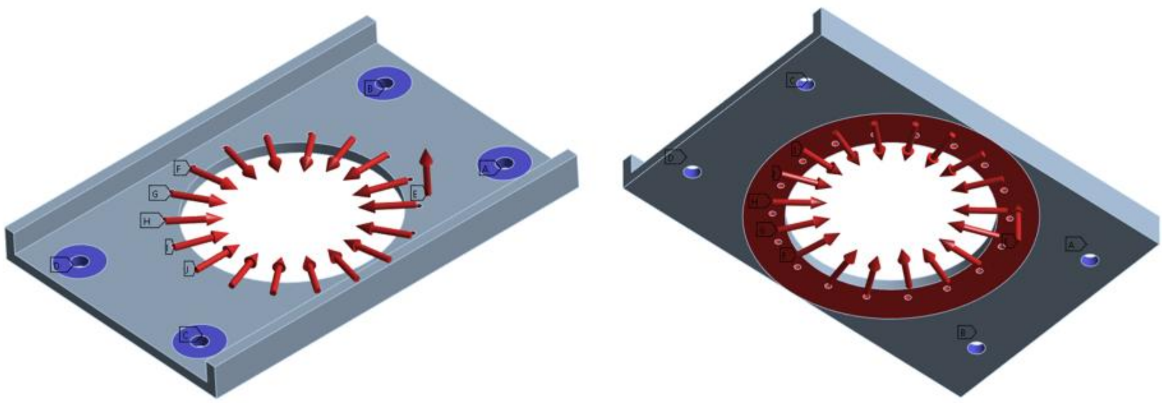


Figure 3: FEA Boundary Conditions used for Structural Simulation

To ensure the computational model accurately represents the physical behaviour of the system, appropriate boundary conditions (BCs) were carefully defined and applied. The following boundary conditions were implemented, along with their corresponding justifications:

1. **Compression-only supports** were applied to the M20 rod holes to represent the contact interaction between the jig and the rods. This condition allows load transfer only when the hole walls are in compression against the rods, accurately capturing the contact stresses that arise as the jig deflects away from the rods.

2. A **frictionless support** was assigned to the washer to constrain displacement in the normal direction while permitting tangential motion. This approach simulates contact between components without introducing artificial normal deformation or frictional resistance.

3. **Reaction forces** were applied at the M8 bolt holes to counterbalance the externally applied loads acting on the mock composite plate. In addition, a **15 kN vertical load** was applied to the underside of the plate to replicate the loading conditions imposed by the Instron testing machine during experimental evaluation.

Finite Element Analysis – Post Process

Deformation:

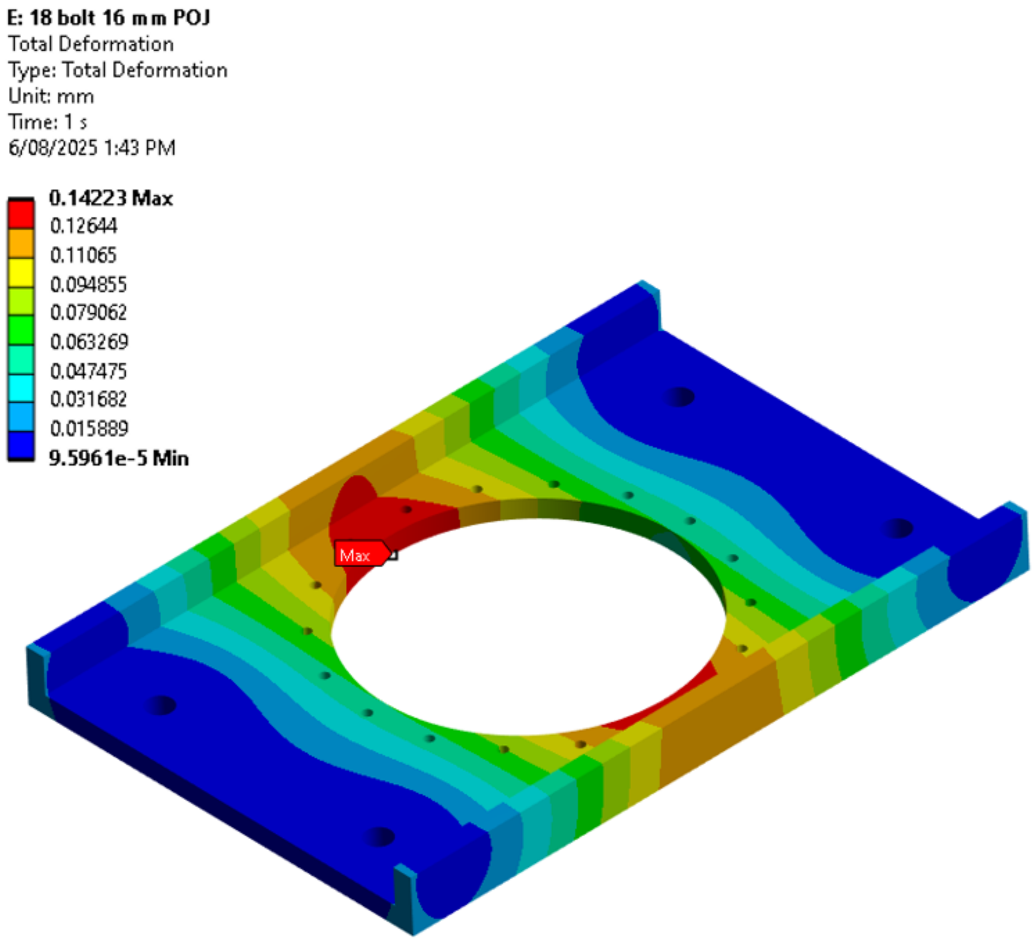


Figure 4: Deformation Contour of the Testing Jig

Figure 4 shows that the largest deformation are located in the central region of the specimen. This behaviour is expected, as the central region is the furthest from the fixture and therefore experiences the greatest bending moment.

Von Mises Stress:

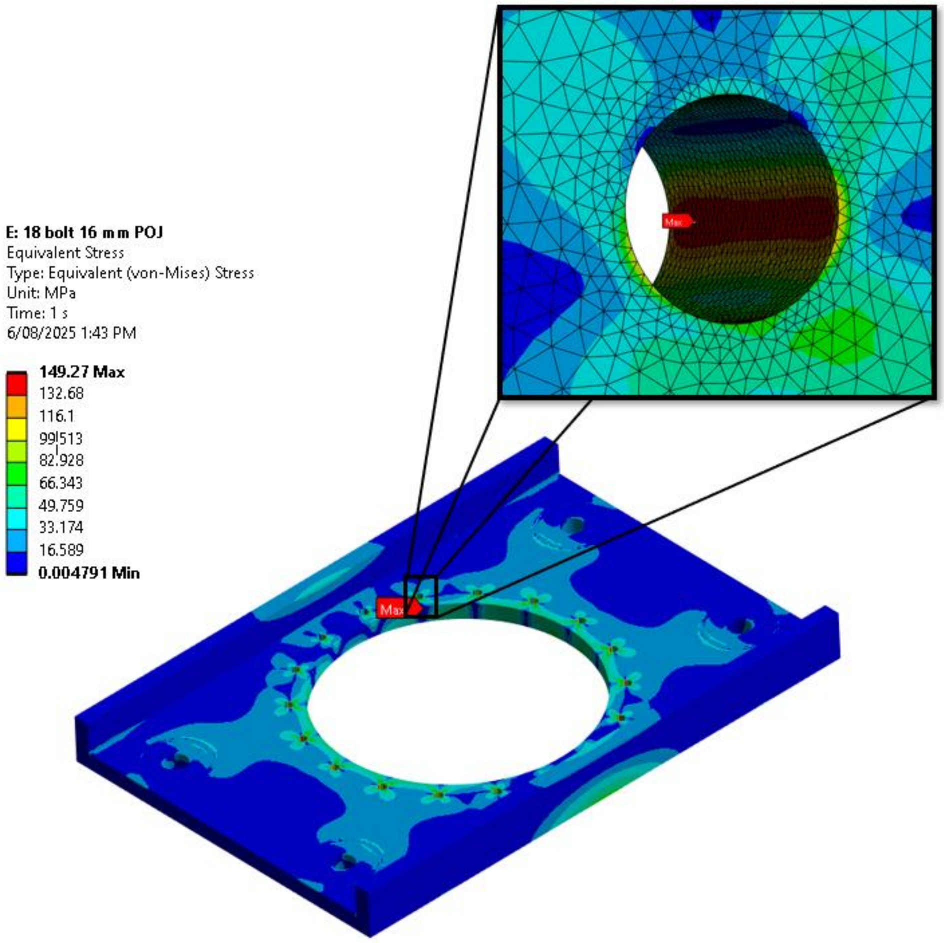


Figure 5: Stress Contour and Concentration of Testing Jig

Figure 5 illustrates the stress distribution across the entire jig. The results indicate localised stress concentrations within the M8 bolt holes. This behaviour is expected, as these regions are primarily responsible for resisting deformation caused by the pull-out forces which results in the plate compression against the bolts.

Results

Following approval of the finite element analysis (FEA) model by a certified professional engineer, the testing jig was released for fabrication through the university machine shop. Upon completion, the jig was immediately commissioned for experimental validation, as shown in Figure 6.

The test results demonstrated that a tensile load of 20 kN was successfully applied through the composite specimen with no observable deformation or measurable deflection of the testing jig, confirming the structural adequacy of the design.

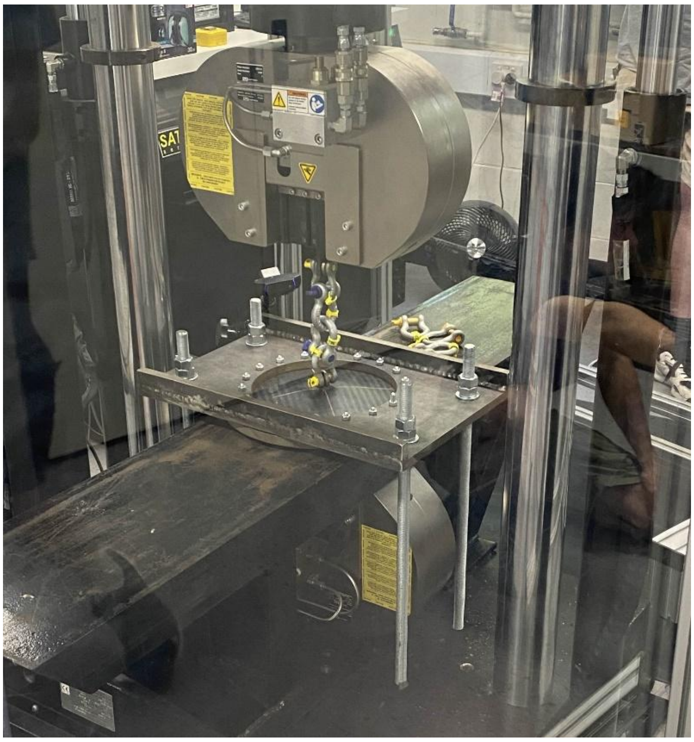
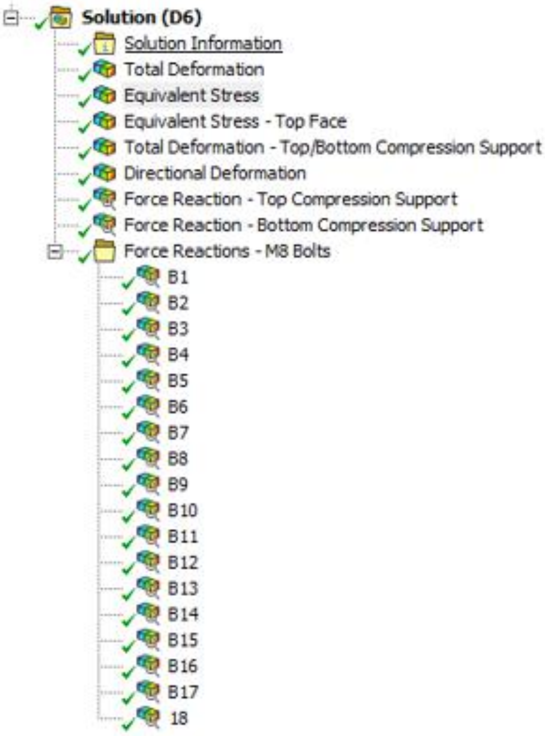
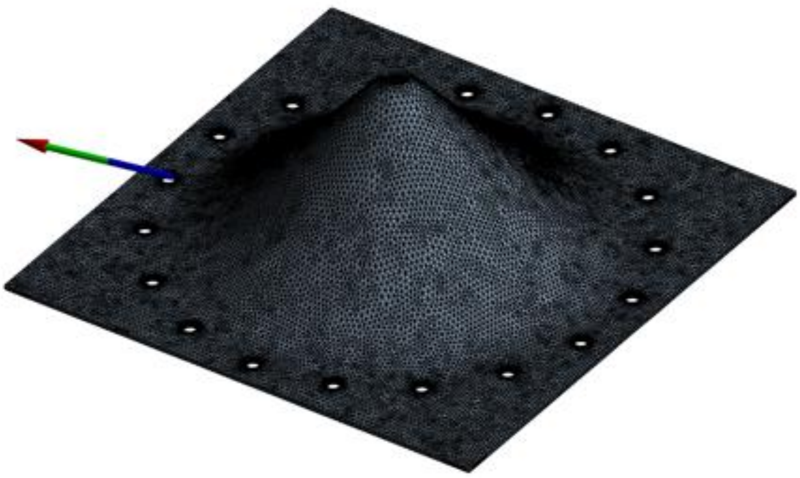


Figure 6: Composite Testing Jig in Use

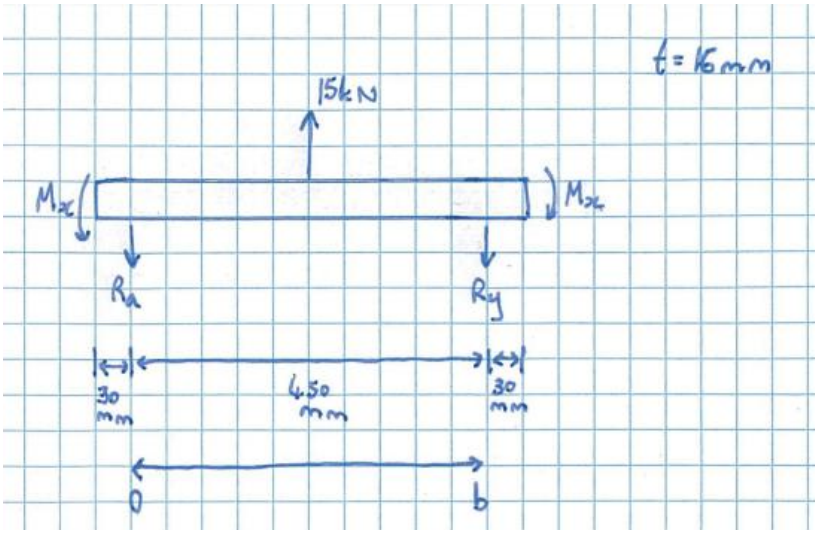
Appendix

Mock Composite Plate Solutions Tree





Hand Calculations



$\sum F_y = 0$

$0 = 15\times10^3 - R_a - R_y$

Deflection @ R_a & $R_b = 0$

Slope @ R_a & $R_b = 0$

$$w\Big|_{y=0} = 0 \quad (1) \qquad \text{and} \qquad w\Big|_{y=b} = 0 \quad (2)$$

$$\frac{\partial w}{\partial y}\Big|_{y=0} = 0 \quad (3) \qquad \text{and} \qquad \frac{\partial w}{\partial y}\Big|_{y=b} = 0 \quad (4)$$

Governing Eq: $\underbrace{\frac{\partial^4 w}{\partial x^2}}_{=0} + 2\underbrace{\frac{\partial^4 w}{\partial x^2 \partial y^2}}_{0} + \frac{\partial^4 w}{\partial y^4} = \frac{P}{D}$

$$\frac{\partial^4 w}{\partial y^4} = \frac{P}{D}$$

$$\frac{\partial^4 w}{\partial y^4} = \frac{P}{D}$$

$$\frac{\partial^3 w}{\partial y^3} = \frac{P}{D} y + C_1$$

$$\frac{\partial^2 w}{\partial y^2} = \frac{1}{2}\frac{P}{D} y^2 + C_1 y + C_2$$

$$\frac{\partial w}{\partial y} = \frac{1}{6}\frac{P}{D} y^3 + \frac{1}{2} C_1 y^2 + C_2 y + C_3$$

$$w = \frac{1}{24}\frac{P}{D} y^4 + \frac{1}{6} C_1 y^3 + \frac{1}{2} C_2 y^2 + C_3 y + C_4$$

Boundary Condition

B.C. ①: When $y = 0$, $w = 0$

$$W = \frac{1}{24}\frac{P}{D} y^4 + \frac{1}{6} C_1 y^3 + \frac{1}{2} C_2 y^2 + C_3 y + C_4$$

$$0 = \frac{1}{24}\frac{P}{D}(0)^4 + \frac{1}{6} C_1 (0)^3 + \frac{1}{2} C_2 (0)^2 + C_3 (0) + C_4$$

$$C_4 = 0$$

B.C. ③: When $y = 0$, $\partial w/\partial y = 0$

$$\frac{\partial w}{\partial y} = \frac{1}{6}\frac{P}{D} y^3 + \frac{1}{2} C_1 y^2 + C_2 y + C_3$$

$$0 = \frac{1}{6}\frac{P}{D} 0^3 + \frac{1}{2} C_1 (0)^2 + C_2 (0) + C_3$$

$$C_3 = 0$$

B.C. ④: When $y = b$, $\partial w/\partial y = 0$

$$\frac{\partial w}{\partial y} = \frac{1}{6}\frac{P}{D} y^3 + \frac{1}{2} C_1 y^2 + C_2 y + \cancelto{0}{C_3}$$

$$0 = \frac{1}{6}\frac{P}{D}(b)^3 + \frac{1}{2} C_1 b^2 + C_2 \times b$$

$$C_2 = -\frac{1}{6}\frac{P}{D} b^2 - \frac{1}{2} C_1 b$$

B.C. ②: When $y = b$, $w = 0$

$$W = \frac{1}{24}\frac{P}{D} y^4 + \frac{1}{6} C_1 y^3 + \frac{1}{2} C_2 y^2 + \cancelto{0}{C_3 y} + \cancelto{0}{C_4}$$

$$0 = \frac{1}{24}\frac{P}{D}(b)^4 + \frac{1}{6} C_1 (b)^3 + \frac{1}{2} C_2 (b)^2$$

Sub in C_2 from previous equation to get C_1

$$0 = \frac{1}{24}\frac{P}{D} b^4 + \frac{1}{6} C_1 b^3 + \frac{1}{2}\left(-\frac{1}{6}\frac{P}{D} b^2 - \frac{1}{2} C_1 b\right)(b)^2$$

$$C_1 = -\frac{1}{2}\frac{P}{D}(b)$$

Sub C_1 into C_2

$$C_2 = -\frac{1}{6}\frac{P}{D} b^2 - \frac{1}{2}\left(-\frac{1}{2}\frac{P}{D} b\right)(b)$$

$$C_2 = \frac{1}{12}\frac{P}{D} b^2$$

Sub C_1, C_2, C_3 & C_4 into w

$$w = \frac{1}{24}\frac{P}{D}y^4 + \frac{1}{6}C_1 y^3 + \frac{1}{2}C_2 y^2 + \cancel{C_3 y}^{0} + \cancel{C_4}^{0}$$

$$w = \frac{1}{24}\frac{P}{D}y^4 + \frac{1}{6}\left(-\frac{1}{2}\frac{P}{D}b\right)y^3 + \frac{1}{2}\left(\frac{1}{12}\frac{P}{D}b^2\right)y^2$$

$$w = \frac{1}{24}\frac{P}{D}\left(y^4 - 2by^3 + b^2y^2\right)$$ - Deflection

M_x and M_y at fixed ends

When $y=0$: $$\frac{\partial^2 w}{\partial y^2} = \frac{1}{2}\frac{P}{D}(0)^2 + \left(-\frac{1}{2}\frac{P}{D}b\right)(0) + \left(\frac{1}{12}\frac{P}{D}b^2\right)$$

$$= \frac{1}{12}\frac{Pb^2}{D}$$

When $y=b$: $$\frac{\partial^2 w}{\partial y^2} = \frac{1}{2}\frac{P}{D}(b)^2 + \left(-\frac{1}{2}\frac{P}{D}b\right)(b) + \frac{1}{12}\frac{Pb^2}{D}$$

$$= \frac{1}{2}\frac{Pb^2}{D} - \frac{1}{2}\frac{Pb^2}{D} + \frac{1}{12}\frac{Pb^2}{D}$$

$$= \frac{1}{12}\frac{Pb^2}{D}$$

M20 Bolt

Moments: $$M_x = -D\left(\cancel{\frac{\partial^2 w}{\partial x^2}}_{0} + \nu\frac{\partial^2 w}{\partial y^2}\right)$$

Assumption (Poisson Ratio of steel)
$\nu = 0.3$

$$= -\cancel{D}\left(0 + \frac{1}{3}\left(\frac{1}{12}\frac{Pb^2}{\cancel{D}}\right)\right)$$

$$= -\frac{1}{36}Pb^2 \rightarrow -\frac{1}{36}\times(15\times10^3)\times(0.45)^2 \rightarrow -84.375 \text{ Nm}$$

$$M_y = -D\left(\frac{\partial^2 w}{\partial y^2} + \nu\cancel{\frac{\partial^2 w}{\partial x^2}}^{0}\right)$$

$$= -\cancel{D}\left(\frac{1}{12}\frac{Pb^2}{\cancel{D}}\right)$$

$E = 200\times10^9$ Pa

$$= -\frac{1}{12}Pb^2 \rightarrow -\frac{1}{12}\times(15\times10^3)\times(0.45)^2 \rightarrow -253.125 \text{ Nm}$$

$y = \boxed{0.51/2}$

Max deflection mid point

$$8.33\times10^{-3}\left(\left(\frac{0.45}{2}\right)^4 - 1.02\times\left(\frac{0.45}{2}\right)^3 + 0.2601\times\left(\frac{0.45}{2}\right)^2\right)$$

$$= 3.43\times10^{-5} \text{ mm}$$

Flex Rigidity

$$D = \frac{Et^3}{12(1-\nu^2)}$$

$$= \frac{200\times10^9\times0.016^3}{12(1-0.3^2)}$$

$$= 75\,018.32$$

$$w = \frac{1}{24}\times\frac{15\times10^3}{75\,018.32}\left(y^4 - 0.51\times2y^3 + 0.51^2y^2\right)$$

$$= 8.33\times10^{-3}\left(y^4 - 1.02y^3 + 0.2601y^2\right)$$